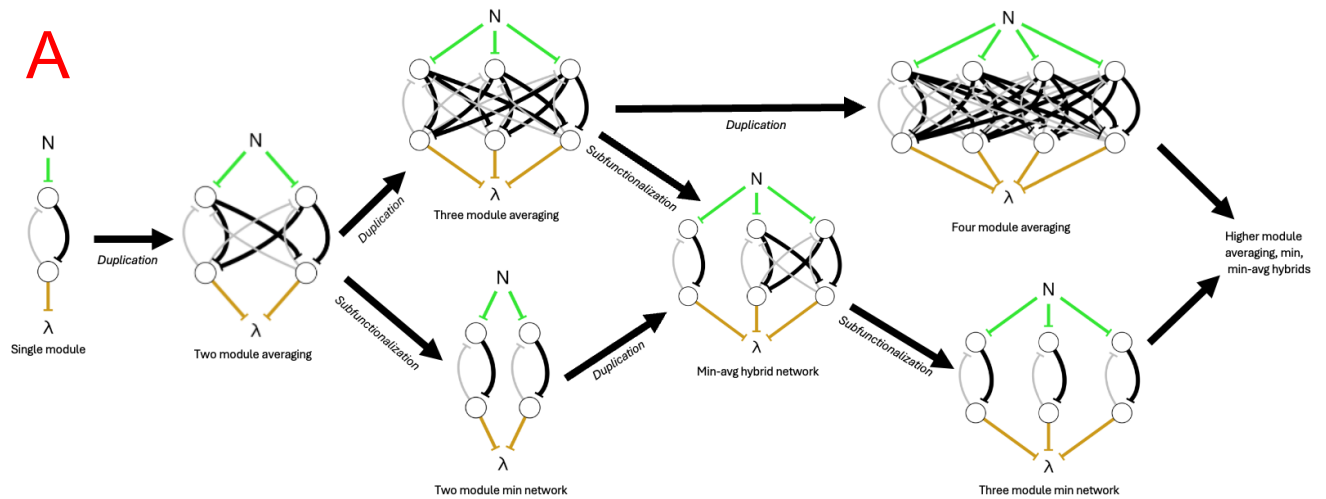
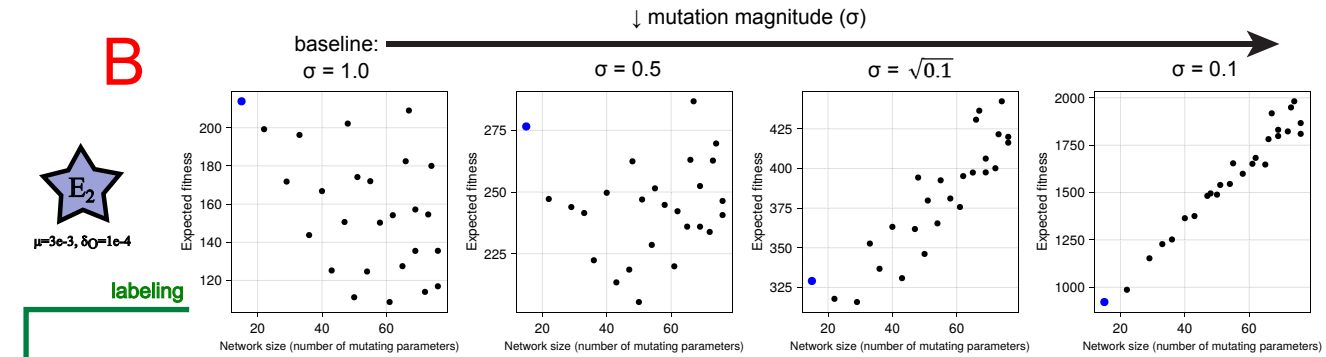
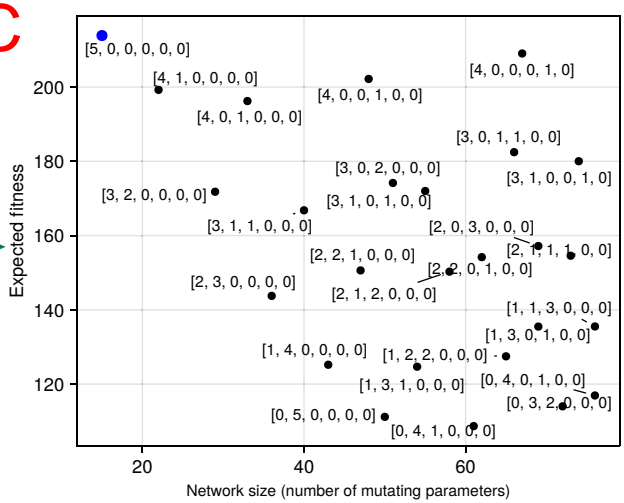
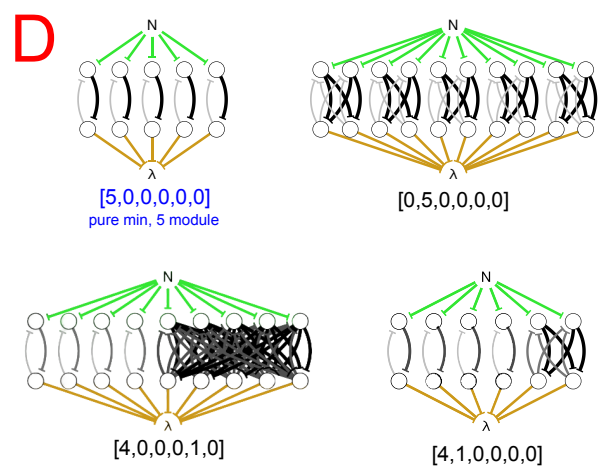

A
N
λ
Single module
Duplication
Two module averaging
Three module averaging
Four module averaging
Higher module averaging, min, min-avg hybrids
Two module min network
Min-avg hybrid network
Three module min network
Subfunctionalization


B
↓ mutation magnitude (σ)
baseline:
σ = 1.0
σ = 0.5
σ = √0.1
σ = 0.1
Expected fitness
Network size (number of mutating parameters)

E2
μ=3e-3, δO=1e-4

labeling


C
[5, 0, 0, 0, 0, 0]
[4, 1, 0, 0, 0, 0]
[4, 0, 1, 0, 0, 0]
[4, 0, 0, 1, 0, 0]
[4, 0, 0, 0, 1, 0]
[3, 2, 0, 0, 0, 0]
[3, 1, 1, 0, 0, 0]
[3, 0, 2, 0, 0, 0]
[3, 1, 0, 1, 0, 0]
[3, 0, 1, 1, 0, 0]
[3, 1, 0, 0, 1, 0]
[2, 3, 0, 0, 0, 0]
[2, 2, 1, 0, 0, 0]
[2, 1, 2, 0, 0, 0]
[2, 2, 0, 1, 0, 0]
[2, 1, 1, 1, 0, 0]
[2, 3, 0, 0, 0, 0]
[1, 4, 0, 0, 0, 0]
[1, 3, 1, 0, 0, 0]
[1, 2, 2, 0, 0, 0]
[1, 3, 0, 1, 0, 0]
[1, 1, 3, 0, 0, 0]
[0, 5, 0, 0, 0, 0]
[0, 4, 1, 0, 0, 0]
[0, 4, 0, 1, 0, 0]
[0, 3, 2, 0, 0, 0]
Expected fitness
Network size (number of mutating parameters)


D
N
λ
[5,0,0,0,0,0]
pure min, 5 module
[0,5,0,0,0,0]
[4,0,0,0,1,0]
[4,1,0,0,0,0]